MATH 121:

Commutative Algebra

Essential Information

Class Meetings (10A):	Tuesday/Thursday 10:10am – 12:00pm
X-Hour:	Friday 3:30pm – 4:20pm
Class Location:	Kemeny 242
Office Hours:	Thursday: 12:00pm - 1:30pm
Instructor:	Juliette Bruce
Instructor Email:	juliette.bruce@dartmouth.edu
Instructor Office:	Kemeny 334

Daily Update

Course Information

Course Description

This course develops the core language and techniques of commutative algebra with an eye toward applications in algebraic geometry and related areas. Topics may include Noetherian rings and modules, localization and primary decomposition, dimension theory, integral dependence, completions, and homological methods (depth, regular sequences, Ext/Tor). Emphasis will be placed on both structural theorems and concrete computations through examples and problem sets. Students should be comfortable with basic ring theory.

Textbooks

There are many excellent textbooks introducing commutative algebra; however, I do not view any of them as perfect fits for this course. As such, I am going to suggest a number of books each with a different flavor:

- *Undergraduate Commutative Algebra* - Miles Reid
- *ommutative Algebra with a View Toward Algebraic Geometry* - David Eisenbud
- *Math 614: Course Notes* - Mel Hochster (available here)
- *Ideals, Varieties, and Algorithms* - Cox, Little, O'Shea

Please choose the source that is best for you. I will do my best to match suggested readings to the sources above. Note the readings assigned are overly inclusive, and you will have to pick and choose what is on topic.

This syllabus was converted from Markdown to PDF using ChatGPT.

Tentative Course Schedule

Week	Date	Topic	Worksheet	Reading	Misc.
1.1	March 31 (Tu)	Basics	pdf / tex	R-0, E-0, H-1.1	
1.2	April 2 (Th)	Prime/Maximal Ideals	pdf / tex/m2	R-1.8, E-1, H-1.4	
1.3	April 3 (F)	Gröbner Bases I	pdf / tex / m2	CLS-2	
2.1	April 7 (Tu)	Zariski Topology I	pdf / tex	R-5, E-1.1	
2.2	April 9 (Th)	Algebraic Sets	pdf / tex	R-5, E-1.1, H-1.1	Quiz 1
2.3	April 10 (F)	Valuation Rings	pdf / tex	R-8, E-11.1, H-(p. 65)	Student Only
3.1	April 14 (Tu)	Gröbner Bases II	pdf / tex	CLS-2	
3.2	April 16 (Th)	Nullstellensatz	pdf / tex	R-5, E-1.6/4.6, H-4.1	Salim
3.3	April 17 (F)	Review	n/a	n/a	Student Only
4.1	April 21 (Tu)	Modules I	pdf / tex	R-(2.1-5), E-2.2, H-(6.1-2)	
4.2	April 23 (Th)	Localization	pdf / tex	R-6, E-2.1, H-(2.3-4)	Quiz 2
4.3	April 24 (F)	Zariski Topology II	pdf / tex	R-(1.10-13), H-(2.4-5)	
5.1	April 28 (Tu)	Review	n/a	n/a	Quiz 3
5.2	April 30 (Th)	Review	n/a	n/a	
5.3	May 1 (F)	Review	n/a	n/a	Student Only
6.1	May 5 (Tu)	Exactness I	pdf / tex	R-(2.9-10), E-(2.1-3), H-(6.1-4)	
6.2	May 7 (Th)	Exactness II	pdf / tex	R-(2.4-10), E-4.1, H-7.3	
6.3	May 8 (F)	Gröbner Bases III	pdf / tex	CLS-2	
7.1	May 12 (Tu)	Integrality I	pdf / tex	R-4, E-(4.2-5, 8), H-3/4	
7.2	May 14 (Th)	Integrality II	pdf / tex	R-4, E-8/13, H-3/4	
7.3	May 15 (F)	Review	n/a	n/a	
8.1	May 19 (Tu)	Depth I	n/a	E-17	
8.2	May 21 (Th)	Depth II	n/a	E-18	
8.3	May 22 (F)	Review	n/a	n/a	
9.1	May 26 (Tu)	Regular Rings	n/a	E-10	
9.2	May 28 (Th)	Resolutions	n/a	E-19/20	
9.3	May 29 (F)	Review	n/a	n/a	Student Only
10.1	June 2 (Tu)	Going Forward	n/a	n/a	n/a

Note: Many of the above worksheets were adapted from or inspired by worksheets written by Karen Smith (University of Michigan), who developed a series of beautiful worksheets in 2019 for her commutative algebra course, which you can find here. Even when not inspired directly by one of these worksheets, my viewpoint on the subject was meaningfully shaped by Professor Smith, for which I am deeply grateful.

Note: The preamble document for the LaTeX worksheets can be found here. As a warning this changes frequently as new worksheets are written.

Course Format

Major Dates:

- May 1, 2026: Last day to tell me your final project topic.
- June 2, 2026: Lab Notebook Due (In-Class)
- June 2, 2026: Final Project Due (In-Class)

Grading:

Your final grade will be calculated on the standard grade scale according to the following breakdown:

25% - Quizzes	**50% - Lab Notebook**	**25% - Final Project**

Quizzes

Throughout the course, there will be short 5-15 minute quizes with the goal of helping us reflect on what we are learning, focus our attention on what areas where we are confused, and to hold ourselves accountable for engaging in the course. The quizzes may be announced or unannounced, and they may cover material found only in the assigned reading. Be sure to pay attention to the Daily Update! Your lowest quiz score will be dropped. Quizzes cannot be made up.

Lab Notebook

Each class period will include a worksheet that we will work on in class in small groups. Everyone is expected to participate and work together in a collaborative setting. These worksheets are intentionally designed to be exploratory and may not include every definition or theorem needed to complete them. The goal is for you to develop the ideas together by experimenting with examples, making conjectures, and discussing possible approaches with your group. These worksheets will not be fully finished in-class.

Each week you should find time to complete the worksheet and type up carefully thought-out solutions to the exercises on the weekly workseeets. These should be in your own words. At the end of the term, you will submit your lab notebook as a single typeset file containing solutions to every worksheet in the course. Your Lab Notebook is due the last day of class: June 2, 2026. Late lab notebooks will not be accepted.

Final Project

For your final project you must select a narrow topic or major theorem in commutative algebra not covered in this course and produce short report on this topic/theorem. The report should include: i) a brief 3-6 page introduction/explanation of the topic/theoremwritten at a level appropriate for another student in the class, ii) a 1-2 page worksheet for the topic/theorem similar to those used in class, and iii) a solution key to your worksheet. All of these materials must be typeset in LaTeX. The final project is due the last day of class: June 2, 2026. Late projects will not be accepted.

Many topics and theorems are appropriate for this, and the goal is to allow you to find something that interests yous. If you are unsure of whether a topic/theorem is approriate please come talk to me. You must tell me what your topic is by May 1, 2026. Here is an entirely random and idiosyncratic list of ideas:

- Gröbner bases & Buchberger's Algorithm
- Regular Sequences, Koszul Complexes & Depth
- Cohen Macaualay Rings
- Auslander-Buchsbaum Formula
- Modules over A Regular Local Ring
- What Makes A Complex Exact?
- Hilbert Functions & Polynomials
- Stanley–Reisner Ideals/Rings
- Semi-group Rings & Ehrhart Theory
- Hensel's Lemma & Henselization
- Rees Algebras and Blowup Algebras
- Zariski–Nagata Theorem on Symbolic Powers
- Gorenstein Rings
- Kunz's Theorem
- Hochster's Theorem
- Artin–Rees Lemma
- Hilbert Birch Theorem

Course Policies

Typesetting & Writing:

With the exception of in-class quizzes, all materials you submit for this class must be typeset in LaTeX. This includes your lab notebook and your final project. Learning to write mathematics clearly in LaTeX is an important professional skill, and part of the goal of this course is to help you become comfortable with it. If you are new to LaTeX, there are numerous online resources that provide good introductions, including: here, here, here, and many mnay more.

Mathematics is a form of writing, and your lab notebook and final project should follow professional best practices. This means writing in full, complete, and grammatically correct sentances. See here and here for some relatively good advice on this topic. Both your lab notebook and final project will be graded, in part, on your writing.

Don't Use Generative AI:

Please do not use generative AI – i.e. basically any and all large language models (LLMs) including ChatGPT, Claude, Gemini, DeepSeek, etc. – in this course. While these can be helpful tools in learning mathematics, especially advanced mathematics, they are often not a panacea and can become a hindrance to learning. Using generative AI or LLMs for either your lab notebook or final project will be considered cheating and a violation of the honor code.

Course Expectations:

I believe in the axioms laid out by Professor Federico Ardila, and I will use them to guide my instruction in this course.

1. Mathematical potential is distributed equally among different groups, irrespective of geographic, demographic, and economic boundaries.
2. Everyone can have joyful, meaningful, and empowering mathematical experiences.
3. Mathematics is a powerful, malleable tool that can be shaped and used differently by various communities to serve their needs.
4. Every student deserves to be treated with dignity and respect.

Using these as guiding principles, I expect students to collaborate with me in developing and maintaining an inclusive learning environment where diversity and individual differences are understood, respected, and recognized as a source of strength. Racism, discrimination, harassment, and bullying will not be tolerated. I expect all participants in this course (students and faculty alike) to treat each other with courtesy and respect.

Mathematics requires active participation. Before each class period, please read the assigned sections. Come to class ready to share what you have learned and also what remains confusing. Class meetings will involve some interactive lecture and other activities in a variety of formats; you will get the most out of each class day if you arrive ready to engage. In all settings, collaborate thoughtfully and ask questions respectfully: everyone should be able to participate.

Honor Principle

We will strictly enforce Dartmouth's Academic Honor Principle.

Additional Information

Mental Health:

TThe academic environment at Dartmouth is challenging; our terms are intensive, and classes are not the only demanding part of your life. There are a number of resources available to you on campus to support your wellness:

- Undergraduate Dean;
- The Counseling Center; and the
- Student Wellness Center.

I encourage you to use these resources to take care of yourself throughout the term. Please come speak to me if you experience any difficulties, or would like help accessing any of these resources.

Student Accessibility Services:

Students requesting disability-related accommodations and services for this course are encouraged to schedule a phone/video meeting with me as early in the term as possible. This conversation will help to establish what supports are built into my online course. In order for accommodations to be authorized, students are required to consult with Student Accessibility Services (SAS; 603-646-9900) and to email me their SAS accommodation form. We will then work together with SAS if accommodations need to be modified based on the online learning environment. If students have questions about whether they are eligible for accommodations, they should contact the SAS office. All inquiries and discussions will remain confidential.

Title IX:

At Dartmouth, we value integrity, responsibility, and respect for the rights and interests of others, all central to our Principles of Community. We are dedicated to establishing and maintaining a safe and inclusive campus where all have equal access to the educational and employment opportunities Dartmouth offers. We strive to promote an environment of sexual respect, safety, and well-being. In its policies and standards, Dartmouth demonstrates unequivocally that sexual assault, gender-based harassment, domestic violence, dating violence, and stalking are not tolerated in our community.

The Sexual Respect Website at Dartmouth provides a wealth of information on your rights with regard to sexual respect and resources that are available to all in our community.

Please note that, as a faculty member, I am obligated to share disclosures regarding conduct under Title IX with Dartmouth's Title IX Coordinator. Confidential resources are also available, and include licensed medical or counseling professionals (e.g., a licensed psychologist), staff members of organizations recognized as rape crisis centers under state law (such as WISE), and ordained clergy (see here).

Should you have any questions, please feel free to contact Dartmouth's Title IX Coordinator or the Deputy Title IX Coordinator for the Guarini School. Their contact information can be found on the sexual respect website.

Religious Observances:

Some students may wish to take part in religious observances that occur during this academic term. If you have a religious observance that conflicts with your participation in the course, please meet with me as soon as possible, or before the end of the second week of the term (at the latest) to discuss appropriate adjustments. Dartmouth has a deep commitment to support students' religious observances and diverse faith practices.